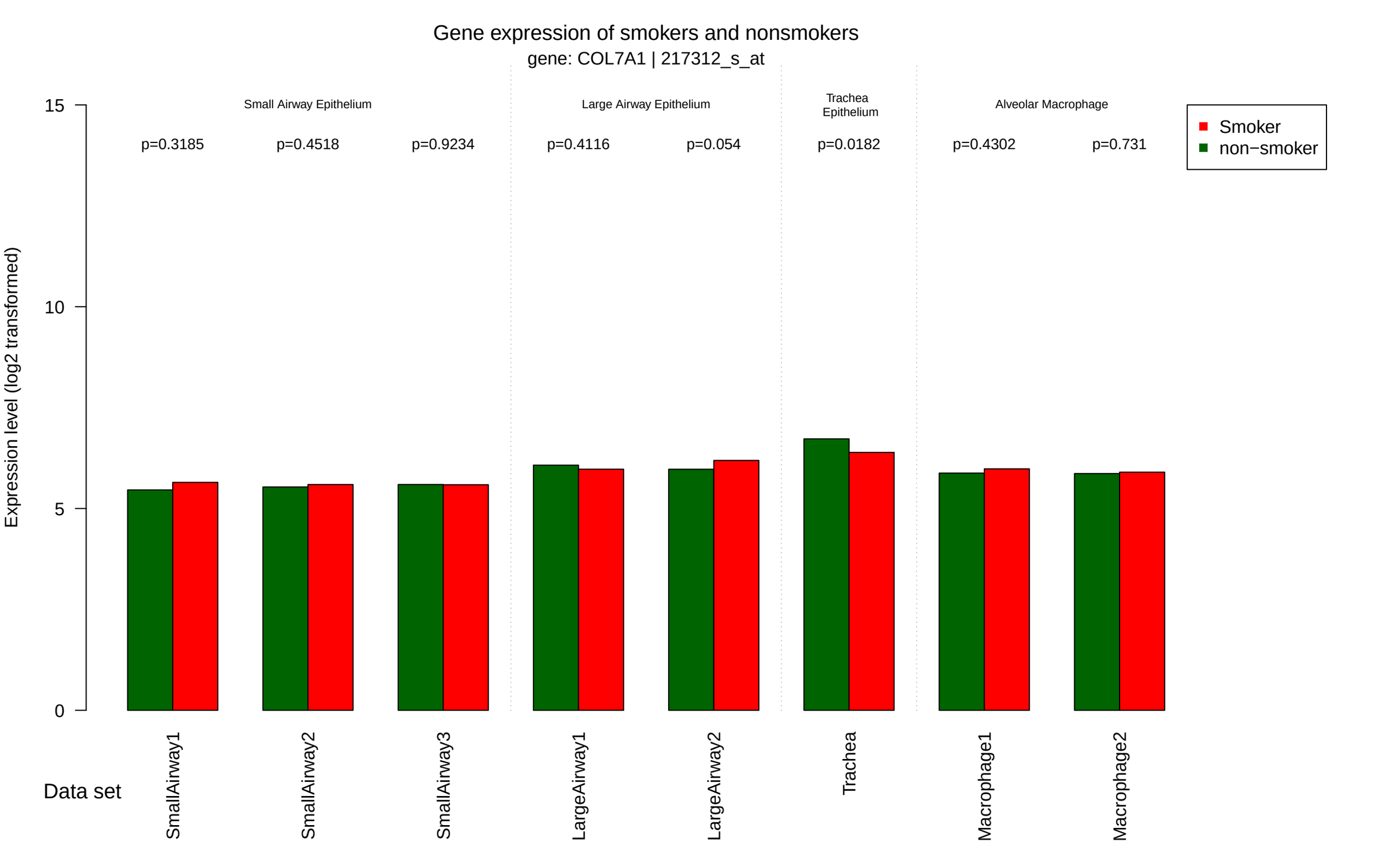
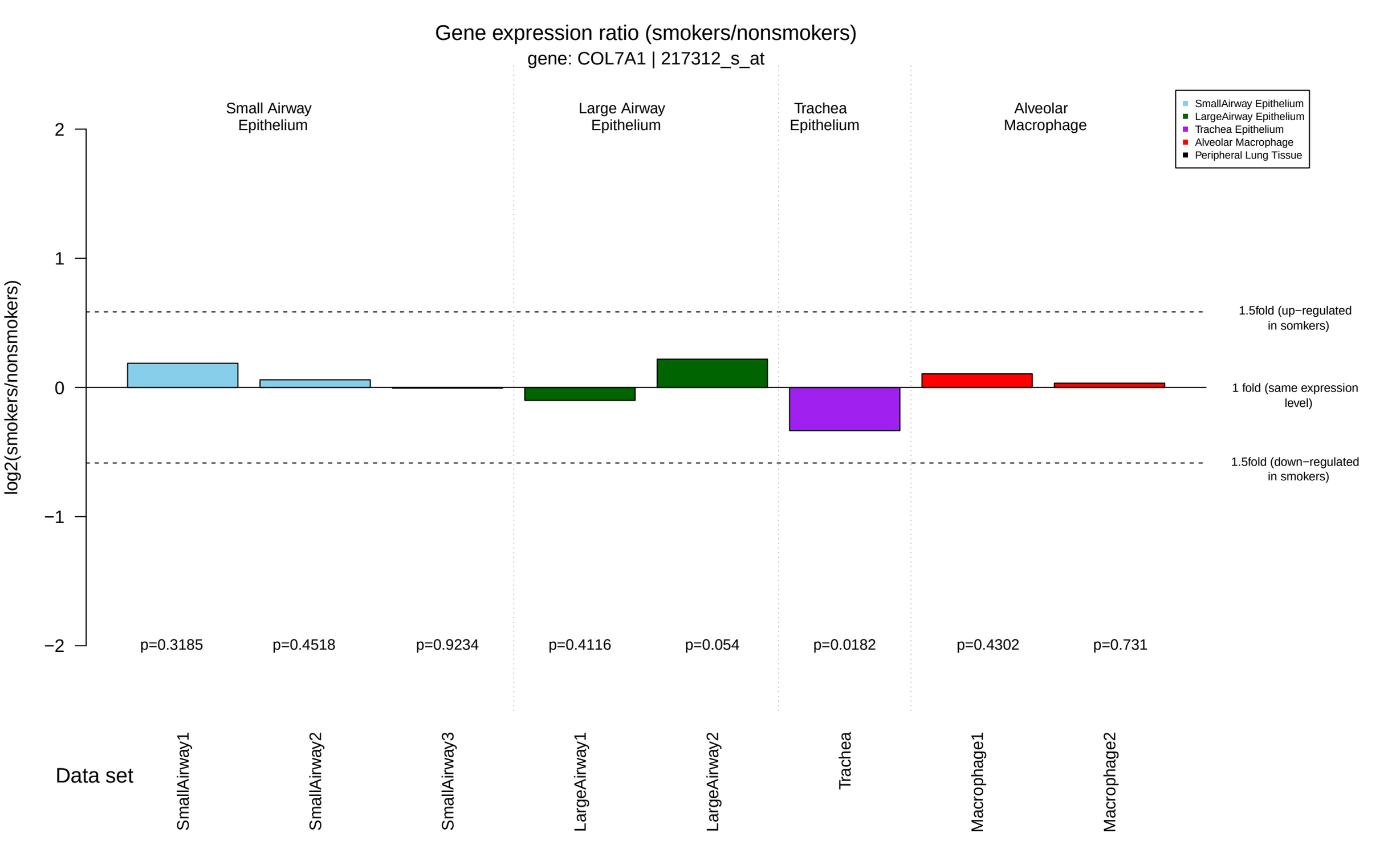
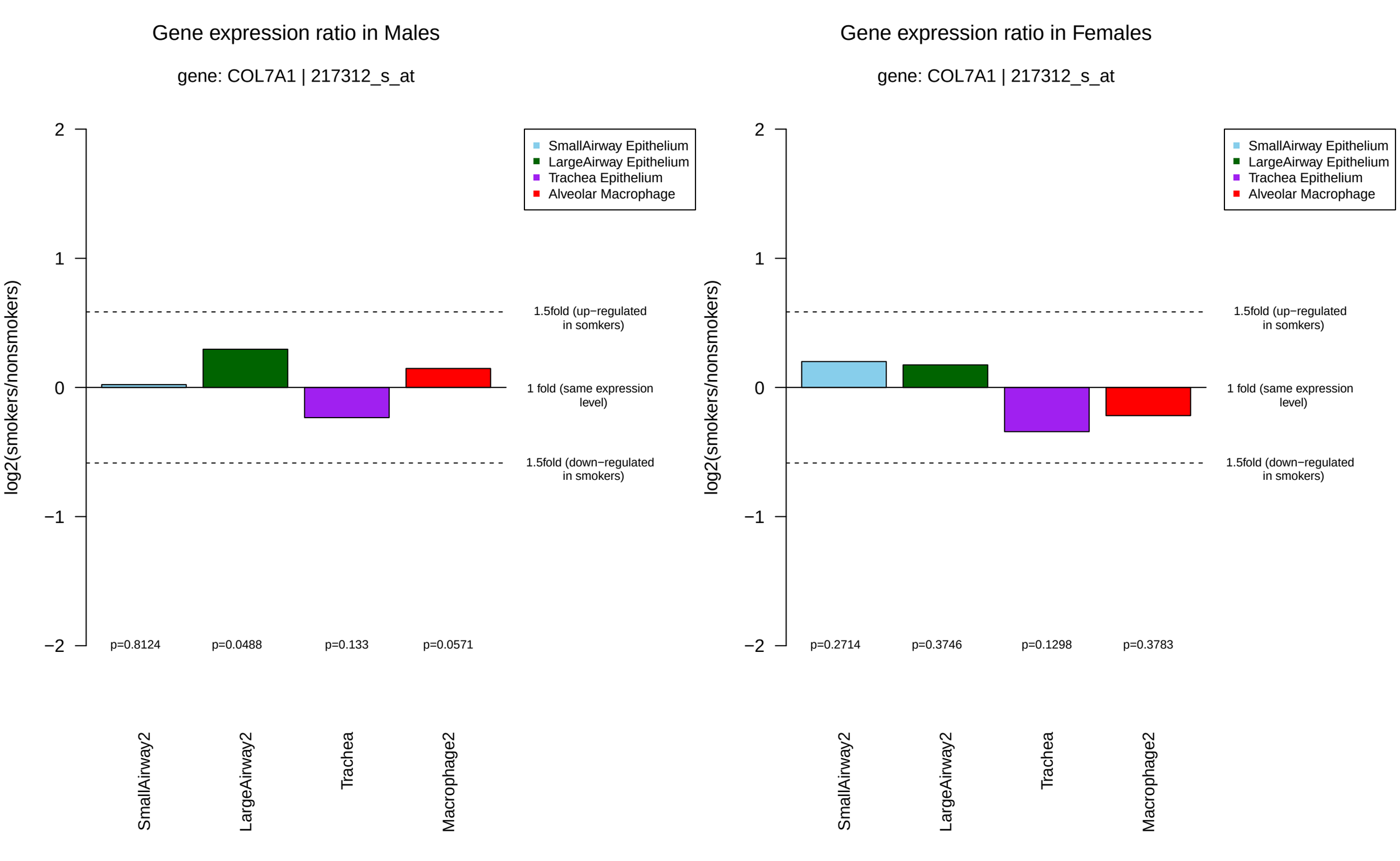
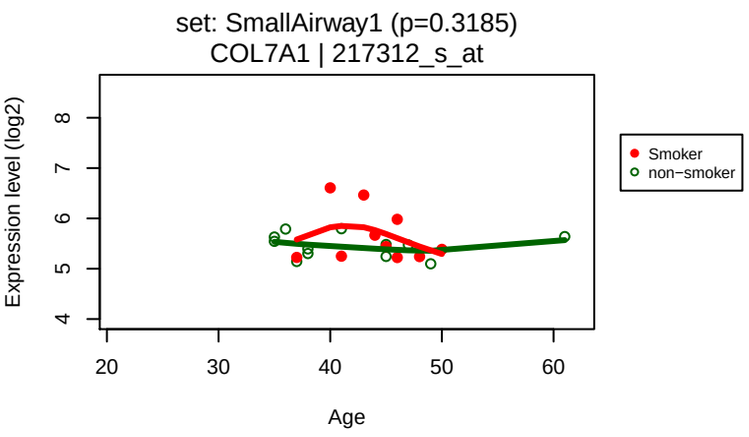
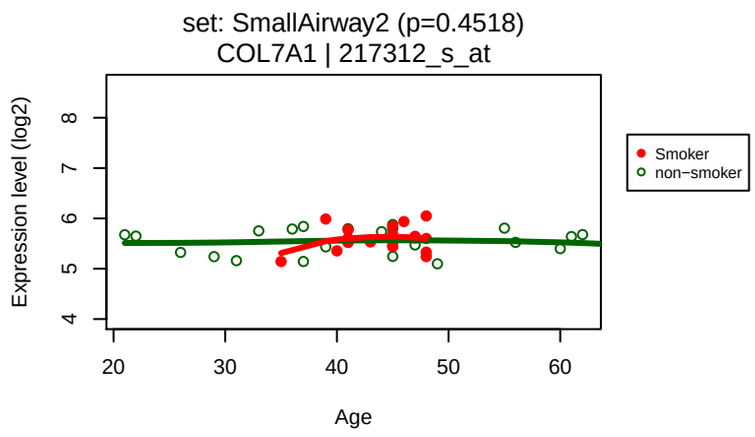
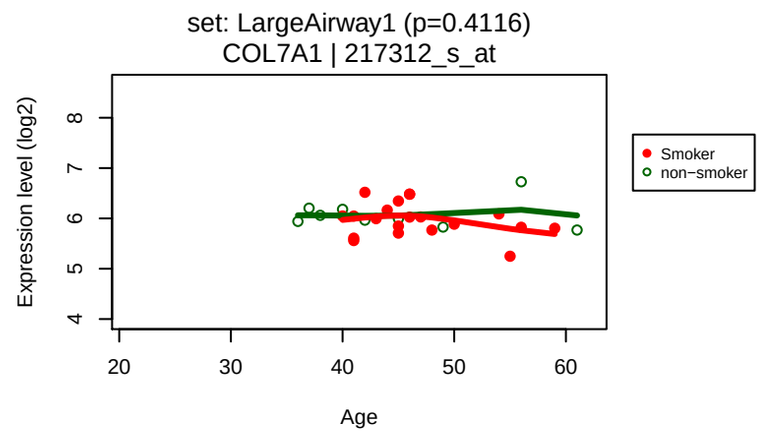

Gene expression of smokers and nonsmokers
gene: COL7A1 | 217312_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3185
p=0.4518
p=0.9234
p=0.4116
p=0.054
p=0.0182
p=0.4302
p=0.731
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

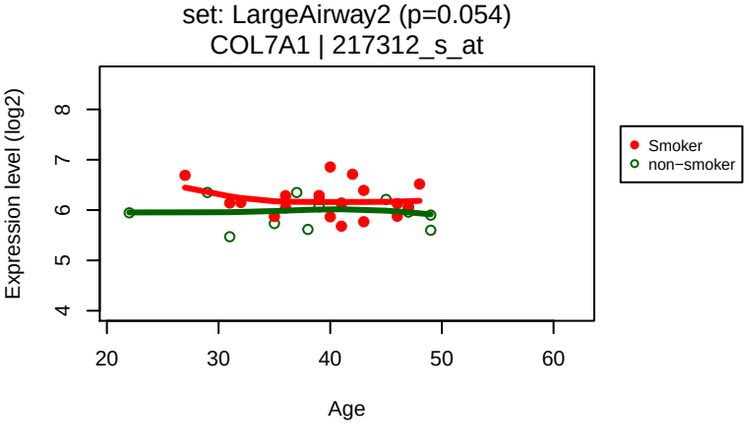

Gene expression ratio (smokers/nonsmokers)
gene: COL7A1 | 217312_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3185
p=0.4518
p=0.9234
p=0.4116
p=0.054
p=0.0182
p=0.4302
p=0.731
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

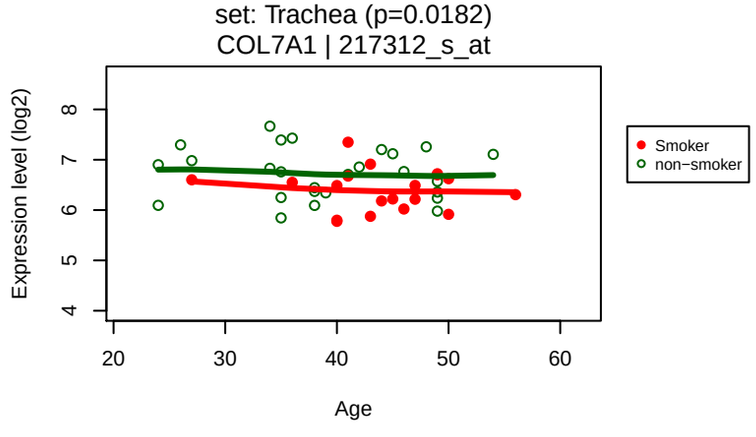

Gene expression ratio in Males
gene: COL7A1 | 217312_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8124
p=0.0488
p=0.133
p=0.0571
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL7A1 | 217312_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2714
p=0.3746
p=0.1298
p=0.3783
SmallAirway2
LargeAirway2
Trachea
Macrophage2

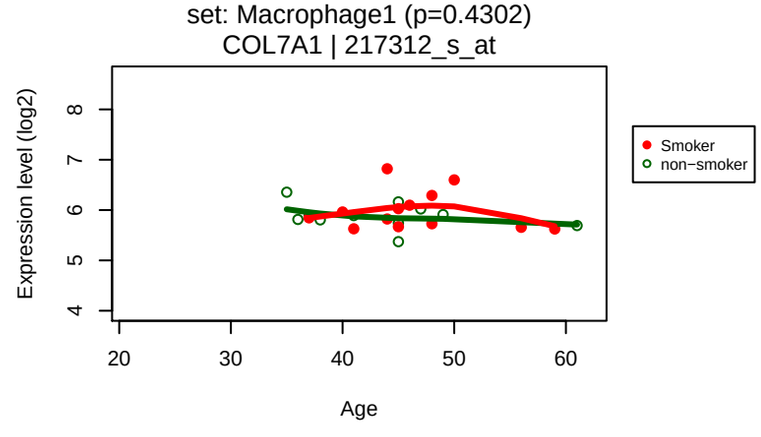

set: SmallAirway1 (p=0.3185)
COL7A1 | 217312_s_at
Expression level (log2)
Age
Smoker
non−smoker

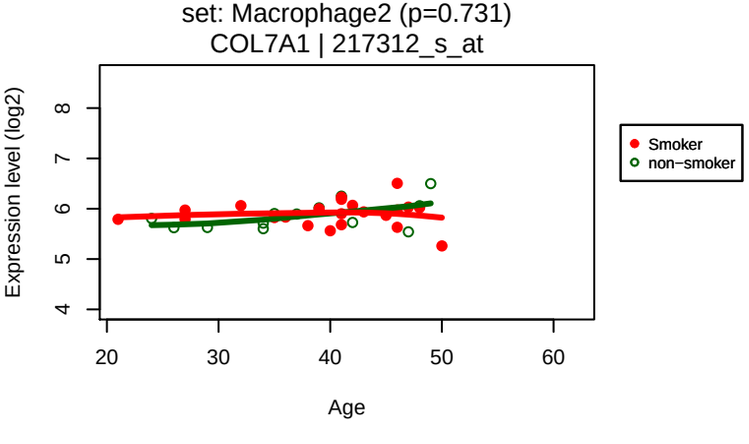

set: SmallAirway2 (p=0.4518)
COL7A1 | 217312_s_at
Expression level (log2)
Age
Smoker
non−smoker

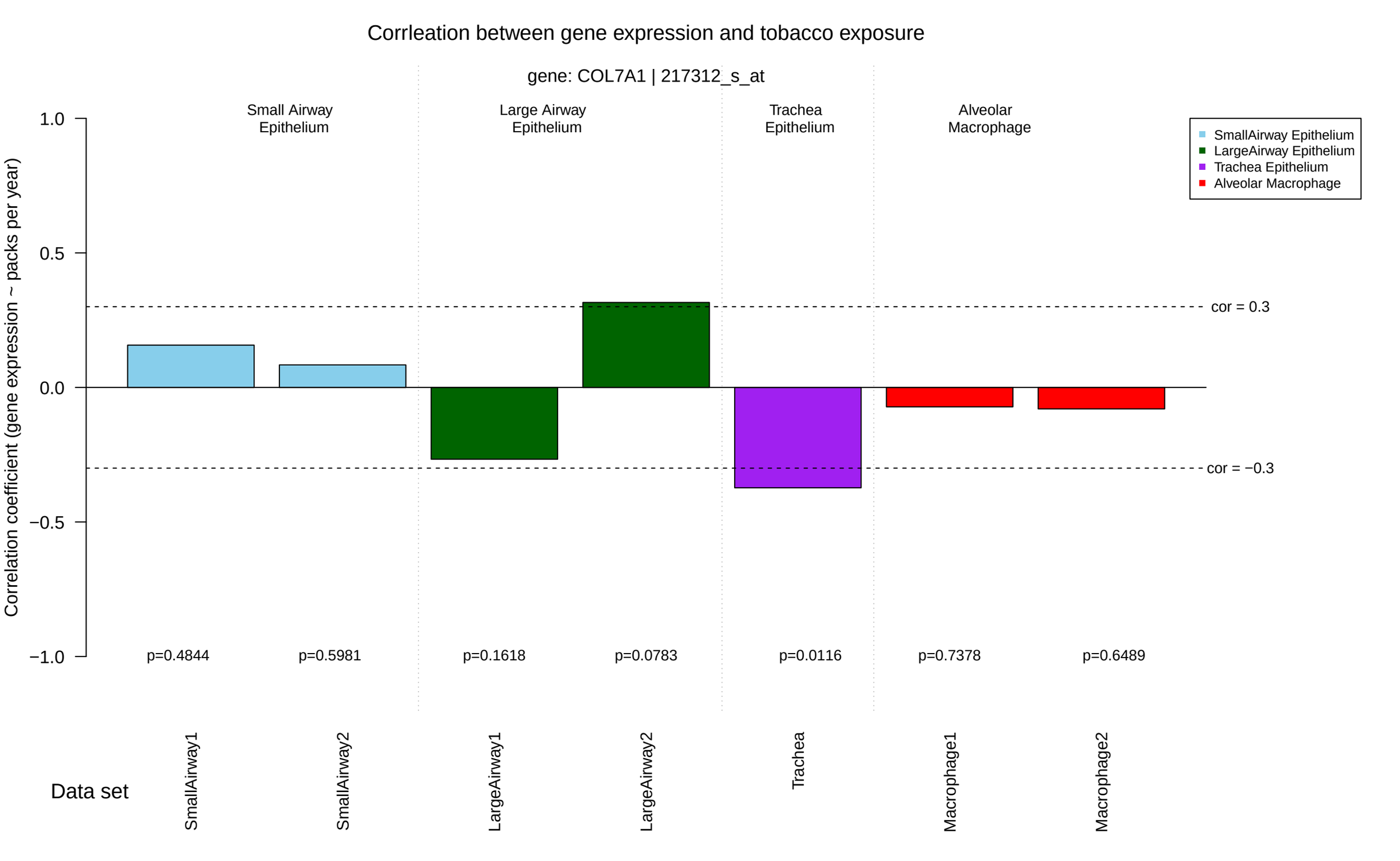

set: LargeAirway1 (p=0.4116)
COL7A1 | 217312_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.054)
COL7A1 | 217312_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0182)
COL7A1 | 217312_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4302)
COL7A1 | 217312_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.731)
COL7A1 | 217312_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL7A1 | 217312_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4844
p=0.5981
p=0.1618
p=0.0783
p=0.0116
p=0.7378
p=0.6489
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2